DENSO
Crafting the Core

NEW

Robot Protective Jacket for Food Processing

Simply fit this jacket over a standard-specification robot to easily and inexpensively automate food manufacturing processes that require cleaning.



Scan here for detailed product information

Compatible robots	VS068/VS087
Compatible controllers	RC8A、RC8

*It can't be used for Communication Interface Flange-A, UL specifications, and Cleanroom type.

Easily fitted to implement low-cost automation of food manufacturing processes

To fit the jacket, simply place it over the robot and tie the drawstrings to hold it in place. Then remove from the robot for cleaning as necessary. Alternatively, the jacket can be secured to a pedestal with a dedicated plate*. It can also be cleaned while fitted to the robot by spraying with water or wiping with a moist cloth.

*Dedicated plate should be supplied by customers.



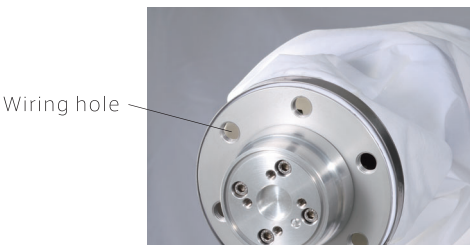
Compatible with chemicals used in food manufacturing processing

The jacket is resistant to a variety of chemicals, ensuring that it will remain clean and sanitary at all times.

Chemicals to which the jacket is resistant
·Sodium hypochlorite aqueous solution (alkaline) ·Sodium hypochlorite pH conditioning liquid (weakly acidic) ·Alcohol ·Hot water (40 to 100 degree C)

Stow wiring by using the dedicated mounting flange

Since cables can be routed from inside the robot protective jacket for food processing through holes in the mounting flange, robot hand cables can be stowed inside the jacket.



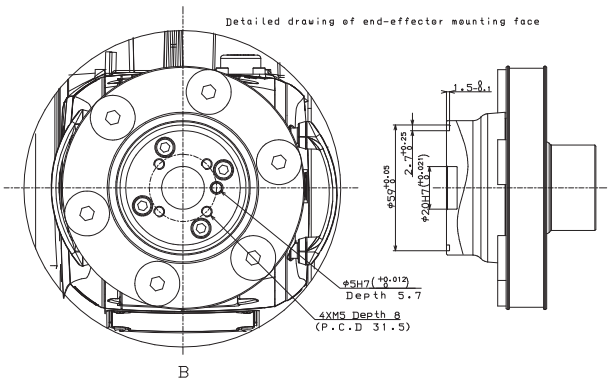
Note: Materials used in parts of this product that may come into contact with food comply with the Specifications and Standards for Food and Food Additives under the Food Sanitation Act (i.e., they meet the requirements set forth in Section 3. Implements, Containers, and Packaging in Ministry of Health and Welfare Notification, No. 370, 1959) (except sewing thread). Materials do not comply with standards in effect in countries outside Japan.

Specifications

Specifications		Unit	VS068		VS087	
			Standard specifications	Robot fitted with jacket	Standard specifications	Robot fitted with jacket
Total arm length (Including No. 1 arm, No. 2 arm and the distance to arm end)		mm	760 (340+340+80)	830 (340+340+150)	955 (445+430+80)	1025 (445+430+150)
Motion angles*2	J1	°	±170	±120*1	±170	±120*1
	J2		+135 ～ -100	+90 ～ -70*1	+135 ～ -100	+90 ～ -70*1
	J3		+153 ～ -120	+140 ～ -20*1	+153 ～ -136	+140 ～ -20*1
	J4		±270	±90*1	±270	±90*1
	J5		±120	+110 ～ -100*1	±120	+110 ～ -100*1
	J6		±360	±240*1	±360	±240*1
Maximum payload		kg	7	6	7	6
Operating temperature range		℃	0～40	0～40*3	0～40	0～40*3
Maximum allowable moment of inertia	J4,J5	kgm2	0.45	0.44	0.45	0.44
Maximum allowable moment	J4,J5	Nm	16.2	14.4	16.2	14.4
	J6		6.86	6.69	6.86	6.69
Signal line and air pipe solenoid valves	Air pipe	–	7 circuits (ø4 × 6, ø6 × 1)	Robot: 7 circuits (ø4 × 6, ø6 × 1) Jacket flange ports: Max 6 circuits (ø4 to ø8) Floor-standing only	7 circuits (ø4 × 6, ø6 × 1)	Robot: 7 circuits (ø4 × 6, ø6 × 1) Jacket flange ports: Max 6 circuits (ø4 to ø8) Floor-standing only
Installation orientation		–	Floor-standing, wall-mounted, ceiling	Floor-standing	Floor-standing, wall-mounted, ceiling	Floor-standing
Weight		kg	49	50	51	52

*1: Range encompasses composite motions for all axes. Single-axis motion conforms to the motion range indicated in the standard specifications except J5.
*2: Varies with the motion range of your robot. You may need to set software limits. *3: Robot temperature will increase more readily when fitted with jacket.

Hand mounting face after jacket installation



* The above illustrations and dimensions are for VS068 / 087.
All six wiring holes are closed with thin head screw plugs.

Robot protective jacket

Model	Type
Robot protective jacket set for food processing (VS068)	For VS068
Robot protective jacket set for food processing (VS087)	For VS087

Options (Must select and use appropriate options)

Model	Application
Cable gland (dia 3 to 5 mm)*1	When hand cable / pipe with a diameter of 4 to less than 6 mm is used
Cable gland (dia 5 to 8 mm)*2	When hand cable / pipe with a diameter of 6 to less than 8 mm is used
Cable gland seal plug (dia 3 to 5 mm)	When part *1 is attached and cable / pipe is not used
Cable gland seal plug (dia 5 to 8 mm)	When part *2 is attached and cable / pipe is not used
Thin head screw plug	When cable /pipe is not used

* The jacket mounting flange must be attached with appropriate cable glands for the robot hand, signal line and pipe used. Close unused wiring holes with plugs.

Also Recommended — For use in strictly controlled hygienic environments, such as food, medical and cosmetic factories

VS050S2

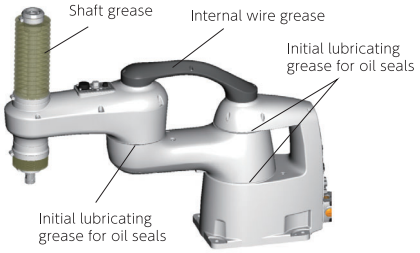
DENSO contributes to automation in food product / medical device / medical product manufacturing and drug preparation. Automation in clean environments prevents the hazards of foreign matter contamination from manual tasks, human error, and operator exposure.



Cleanliness	ISO Class 5
Protection level	Wrist IP67 / Unit IP65

HSR series (NSF H1 grease type)

Quick Accelaration, runs continuously at maximum speed, and stops precisely. The HSR Series that is pursued the basic performances and achieve the true high-speed performance is now compatible with an NSF H1 grease.



Protection level	IP65

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P-FRJ-E 201905